

Once we discard the notion of AI as being that which passes the Turing Test and open the field up to weak AI, a lot of modern mechanical systems fall into the sphere of intelligence. Logic, after all, has been programmed into several computers before. But it's the introduction of "fuzzy logic" that started making the machines "intelligent".

Fuzzy Logic

Fuzzy Logic is a type of reasoning that stems out of the limitations of Boolean Logic. Boolean logic is the idea of extremes – that every sentence can be classified as "completely true" or "completely false". This type of extreme classification was something that most science could get behind because of the two values of 0 and 1 inherent in the binary system. But this hard logic doesn't account for "grey areas" that humans are used to identifying.

In order to make logic more all-encompassing, Dr. Lotfi Zadeh of the University of California at Berkeley came up with the idea of fuzzy logic. To put it in a different way, fuzzy logic is a system intelligent enough to account for degrees between two variables.

A common example used to explain fuzzy logic is with automatic transition gear boxes in cars. Suppose your car is driving along at 30 kmph. If it's a manual gear box, then on a flat road, you might think of putting it in the third gear. If your speed drops a little, you could still keep it in the gear till you accelerate back to your desired speed. Now imagine this was an automatic transition car. If the automatic transition was set to go into the third gear at 30 kmph, then it would do so regardless of the kind of road you were on – and on an



Lotfi Zadeh, the man behind Fuzzy Logic